KSME

Springer

Journal of Mechanical Science and Technology 39 (12) 2025

Original Article
DOI 10.1007/s12206-025-2502-5

Keywords:
· Convolutional neural networks
· Deep learning
· Defect classification
· Image resolution
· Inference time
· Manufacturing
· Water transfer printing

Correspondence to:
Kunsik An
kunsik1214@sejong.ac.kr

# Thease authors equally contributed to this work as first authors.

Citation:
Lee, G., Ndikumana, J., Choi, H., Choi, S., Yang, J., An, K. (2025). Balancing accuracy and efficiency: comparative analysis of deep learning models for defect classification in water transfer printing. Journal of Mechanical Science and Technology 39 (12) (2025) 7201~7210. https://doi.org/10.1007/s12206-025-2502-5

Received August 31st, 2025
Revised September 24th, 2025
Accepted September 24th, 2025

† This paper was presented at ICMDT2025, Arcrea HIMEJI, Himeji, Japan, April 23-25, 2025. Recommended by Editor Hae-Jin Choi

© The Korean Society of Mechanical Engineers and Springer-Verlag GmbH Germany, part of Springer Nature 2025

Balancing accuracy and efficiency: comparative analysis of deep learning models for defect classification in water transfer printing

Gyuseok Lee[1,#], Joel Ndikumana[2,#], Hyunjoon Choi[1], Seonghyun Choi[2], Jaeseok Yang[1] and Kunsik An[2,3]

[1]Department of Mechatronics Engineering, Konkuk University Glocal Campus, 268 Chungwon-daero, Chungju 27478, Republic of Korea, [2]Department of Mechanical Engineering, Sejong University, 209 Neungdong-ro, Gwangjin-gu, Seoul 05006, Republic of Korea, [3]Department of Aerospace System Engineering, Sejong University, 209 Neungdong-ro, Gwangjin-gu, Seoul 05006, Republic of Korea

Abstract This study presents a comparative evaluation of three deep learning models: MobileNetV2, EfficientNetB0, and ResNet152-for defect classification in water transfer printing manufacturing. We assess their performance across six image resolutions (64 × 64 to 600 × 600 pixels), analyzing key metrics such as accuracy, inference time, and computational efficiency. The models were tested on five defect categories: entrapped air, film creasing, film delamination, pattern distortion, and uniform transfer. MobileNetV2 demonstrated the fastest inference times, ranging from 9.6 ms at 64 × 64 resolution to 572 ms at 600 × 600, with test accuracy peaking at 77.13 % (224 × 224) before declining at higher resolutions. EfficientNetB0 offered a strong balance, achieving up to 87.89 % test accuracy at higher resolutions, with inference times increasing from 14 ms (64 × 64) to 823 ms (600 × 600). ResNet152 delivered the highest test accuracy (up to 95.94 % at 600 × 600), but incurred the highest computational cost, with inference times rising from 55.91 ms (64 × 64) to nearly 3 seconds (2979.75 ms) at maximum resolution. These results confirm a non-linear relationship between image resolution and model performance: as resolution increases from 64 × 64 to 600 × 600 pixels, test accuracy improves by 22–53 % depending on the model, while inference times escalate dramatically-by over 5100 % to nearly 5900 % for all architectures.

1. Introduction

Water transfer printing, also known as hydrographic printing or immersion printing, has earned substantial attention due to its versatility, efficiency, and ability to produce high-quality finishes. Unlike traditional printing methods such as screen or pad printing, this technique can accommodate irregular shapes and curved surfaces which makes it ideal for a wide range of applications. Industries such as automotive, aviation, consumer electronics, organic and metallic sensors, and sporting goods have widely adopted this technology to enhance both the aesthetic and functional value of their products [1-5]. For instance, automotive interiors often feature wood grain or carbon fiber finishes achieved through water transfer printing, while camouflage patterns are commonly applied to hunting gear and outdoor equipment using this technique. In electronics, water-assisted transfer printing methods are employed to integrate electronic devices into flexible substrates without defects [6]. The process ability to create durable, high-resolution patterns has made it indispensable for both decorative and functional purposes.

One of the key advantages of water transfer printing is its compatibility with a variety of substrates, including plastics, metals, ceramics, and glass. This flexibility allows manufacturers to apply customized designs across diverse materials without compromising durability or adhesion. Additionally, the process is environmentally friendly compared to other surface treatment

methods because it minimizes waste and supports the use of eco-friendly inks [7]. The cost-effectiveness and scalability of this method make it suitable for both small-scale prototyping and large-scale industrial production. These attributes have positioned water transfer printing as a preferred choice in industries seeking efficient and sustainable solutions for surface decoration.

Polyvinyl alcohol (PVA) has emerged as a revolutionary substrate in water transfer printing (WTP), transforming decoration and fabrication processes across multiple industries. PVA is a water-soluble polymer known for its excellent printability, environmental friendliness, and adaptability to various shapes [6, 7]. During the water transfer process, the PVA film dissolves uniformly upon activation in a water bath, leaving behind the ink pattern ready for transfer onto the substrate. The unique properties of PVA enable precise pattern application even on highly curved or irregular surfaces. Recent advancements in PVA film technology have further enhanced its mechanical strength and reduced distortion during the transfer process [8, 9]. These improvements not only ensure higher-quality results but also expand the potential applications of water transfer printing in emerging fields such as biomedical devices and advanced manufacturing [6, 9, 10]. Their flexibility and compatibility with various printing techniques, such as gravure and UV inkjet printing, make them versatile for producing high-resolution patterns on diverse materials like plastics, metals, and ceramics [11].

The quality assessment and classification of water transfer printing films traditionally relies on expert visual inspection by trained professionals or basic microscopic image analysis, which is time-consuming, subjective, and prone to human error. While some advanced operations employ inspection cameras, these systems typically detect only obvious defects and lack the sophisticated pattern recognition capabilities needed for nuanced quality assessment.

Computational approaches have been explored to address challenges such as texture alignment on complex 3D surfaces in water transfer printing. For instance, computational hydrographic printing methods have demonstrated the ability to simulate film distortion and optimize texture registration, enabling precise alignment during the immersion process [12, 13]. However, these advancements have primarily focused on improving the printing process itself rather than addressing quality control challenges. While simulations can predict how the PVA color film is distorted during printing, they face several limitations [13]. First, simulations struggle with accurately modeling the physical behavior of the film, particularly when dealing with complex geometries or when the film tears or deforms excessively. Additionally, simulations require extensive 3D surface mapping and parameter calibration, which can be time-consuming and may not fully capture the nuances of real-world manufacturing conditions.

Meanwhile, the integration of deep learning in manufacturing has revolutionized quality control processes, enabling automated defect detection and classification with unprecedented accuracy [14, 15]. CNN models have been increasingly applied to materials science and microscopy, accelerating research and quality control through automatic image analysis [16-20]. For instance, in the field of materials science, CNNs have been utilized for the automatic classification of microstructural features in steel, enhancing the understanding of material properties and aiding in quality control [21]. Similarly, in the realm of microscopy, deep learning techniques have been employed to analyze images of microstructures, providing insights into material behavior and defects [19, 20]. Moreover, in the manufacturing of semiconductors, deep learning models have been used to identify defects in wafers, notably improving the yield and reliability of the production process [22]. These applications demonstrate the versatility and effectiveness of deep learning in enhancing quality control across various manufacturing sectors.

While high accuracy has been the primary focus in multi-class classification frameworks, practical manufacturing applications require balancing accuracy with inference time. Inference time impacts power consumption, resource utilization, and latency are critical considerations in industrial environments where efficiency and cost-effectiveness are paramount [23].

The development of lightweight and efficient deep learning models capable of delivering high performance while minimizing computational overhead and maximizing processing speed is critical for real-world applications, particularly in resource-constrained environments such as embedded microscopy systems or portable diagnostic tools. Image quality significantly impacts classification accuracy and is governed by factors like blur, contrast, compression artifacts, and resolution [24, 25]. Reducing resolution inherently degrades spatial information by suppressing fine-grained textures and edges, which are critical for distinguishing microscale structural patterns in films. Image resolution plays a crucial role in deep learning-based image classification, directly affecting both the information available to the model and the computational resources required for processing. Higher resolution images preserve more fine-grained details that may be essential for distinguishing between similar classes, but they also increase memory requirements and computational load, potentially leading to longer training and inference times [26].

The relationship between image resolution and classification performance is not always straightforward. While intuition suggests that higher resolution should lead to better performance, this is not universally true. Higher resolutions can preserve finer details but often lead to increased computational demands and may result in diminishing returns or overfitting. A notable study using the HyperKvasir dataset assessed the performance of ResNet-152 and DenseNet-161 for endoscopic image classification across resolutions from 32 × 32 to 512 × 512 pixels. Both models demonstrated improved performance at the highest resolution, consistent with the principle that higher resolutions preserve more details crucial for accurate classification, particularly in applications where subtle features are important [25].

Another study investigated the impact of image resolution on the performance of MobileNet and DenseNet121 models for breast cancer diagnosis using ultrasound images. The results showed that MobileNet performed best at 320 × 320 pixels,

achieving an AUC of 0.892, while DenseNet121 performed optimally at 448 × 448 pixels with an AUC of 0.886. The study highlights that higher resolutions generally improve diagnostic accuracy, but the optimal resolution varies by model [27]. Similarly, research on how image resolution affects the performance of CNNs in chest radiograph diagnosis evaluated ResNet34 and DenseNet121 across various resolutions, finding that most diagnostic labels achieved optimal performance at resolutions around 448 × 448 pixels. Notably, pulmonary nodule detection benefited significantly from higher resolutions compared to thoracic mass detection, underscoring the importance of selecting appropriate image resolution for specific applications to optimize CNN performance [28].

This study addresses the critical research gap in deep learning-based defect classification for water transfer printing by systematically evaluating the performance trade-offs between accuracy, efficiency, and computational demands across multiple architectures and image resolutions. We present a comprehensive comparison of three state-of-the-art deep learning models—MobileNetV2, EfficientNetB0, and ResNet152—across six image resolutions ranging from 64 × 64 to 600 × 600 pixels. The Materials and Methods section details our experimental setup, dataset preparation, and model implementation procedures. In Sec. 3, we analyze the performance metrics of each architecture, quantify the resolution-dependent accuracy-efficiency trade-offs on manufacturing constraints. Finally, the Conclusions section synthesizes our findings on the non-linear relationship between image resolution and classification performance, providing practical guidelines for implementing deep learning-based quality control systems in water transfer printing production environments. This systematic evaluation enables manufacturers to select the optimal architecture and resolution configuration based on their specific operational requirements and resource constraints.

2. Materials and methods

2.1 Fabrication details

The experimental procedure for water transfer printing using a commercially available PVA carbon pattern film involves several sequential steps. First, the carbon pattern film is prepared by carefully cutting it to the desired dimensions. Next, a water bath is set up, and its temperature is precisely controlled and maintained at 36.5 °C to ensure optimal conditions for the transfer process. Once the water bath has reached the target temperature, the prepared PVA carbon pattern film is gently placed onto the surface of the water, allowing it to float freely without disturbance. Meanwhile, an activator solution is prepared in a stainless-steel container. This solution is essential for promoting the adhesion of the carbon pattern to the substrate.

After the film has settled on the water bath, the activator solution is uniformly sprayed over the surface of the floating film. This step ensures that the film becomes sufficiently pliable and adhesive for the transfer. Finally, the substrate is carefully submerged through the floating film. This action enables the carbon pattern to adhere and conform to the surface of the substrate, resulting in a successful water transfer and the formation of the final patterned sample.

The water transfer printing process resulted in five distinct categories of surface outcomes on plastic spheres, as illustrated in Fig. 1. These categories were classified based on their visible characteristics. The “entrapped air defects” category is characterized by the presence of small air bubbles trapped between the PVA film and the sphere surface, disrupting the uniformity of the transfer. The “film creasing defects” category occurs when the PVA film folds onto itself during the transfer process, creating visible creases or overlaps on the sphere surface. The “film delamination” category is identified by areas where the PVA film

Table 1. Classification of surface outcomes from water transfer printing based on visible characteristics.

Class name	Surface shape (visible characteristics)
Entrapped air defects	Small air bubbles visible on the surface disrupting uniformity
Film creasing defects	Creases or overlaps due to folding of the film.
Film delamination	Areas where the film has detached, leaving exposed surfaces
Pattern distortion	Distorted or thinned pattern due to excessive stretching
Uniform transfer	Clean and uniform surface with no visible defects

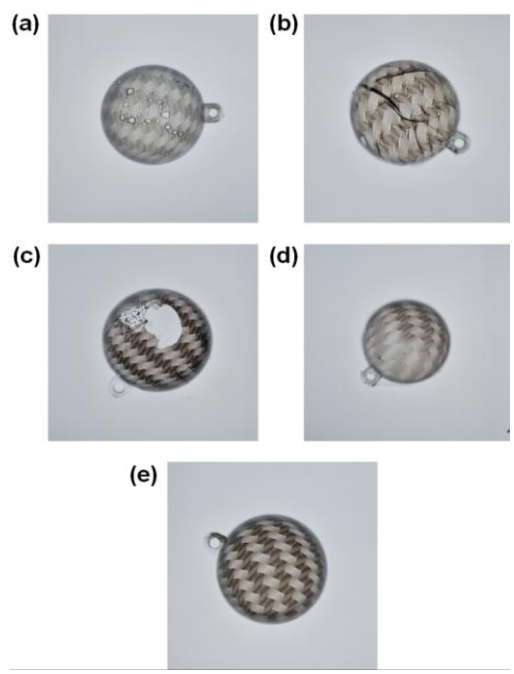

Fig. 1. Visual (original images) representation of the five resulting categories of water-transferred carbon film on plastic spheres: (a) entrapped air defects; (b) film creasing defects; (c) film delamination; (d) pattern distortion; (e) uniform transfer.

has detached or peeled away from the sphere, leaving exposed regions without a proper pattern. The “pattern distortion” category is observed when the PVA film becomes overly stretched during the transfer process, causing distortion or thinning of the carbon pattern. Finally, the “uniform transfer” category represents an ideal outcome where the surface is clean and uniform, with no visible defects. These classifications are critical for evaluating water transfer quality and were used to train a convolutional artificial neural network model for automated recognition and differentiation of these results.

2.2 Methodology workflow

The overall workflow for this study is visually summarized in Fig. 2, which illustrates each step from image data collection and labeling to preprocessing, model building, and performance evaluation. This figure provides a clear overview of the sequential processes and highlights key considerations such as memory usage across varying image resolutions. The following subsections describe each component of the workflow in detail.

Image data collection and labeling

The image data used in this study were acquired at an original resolution of 1689 × 1689 pixels using an optical microscope (Nikon Eclipse LV100ND), enabling high-resolution capture of material surface features. Images were systematically collected across five categories: (a) entrapped air defects; (b) film creasing defects; (c) film delamination; (d) pattern distortion; and (e) uniform transfer. For each category, images were stored in dedicated directories, ensuring clear organization and traceability. The dataset was then labeled programmatically, with each image assigned a class label corresponding to its defect type. The entire dataset was subsequently normalized by scaling pixel values to the range. Ultimately, the dataset was partitioned into training (70 %), validation (15 %), and testing (15 %) subsets through stratified sampling to preserve class distribution across splits, ensuring robust and unbiased model evaluation. performance assessment.

Image preprocessing

Image preprocessing is a essential step in any computer vision workflow, as it directly impacts the quality and consistency of the data fed into machine learning models. Even with high-quality image acquisition, raw images may contain variations in size, orientation, lighting, or noise that can hinder model performance. Preprocessing addresses these issues by standardizing the dataset, ensuring that each image is presented in a uniform format suitable for analysis. This process not only increases the reliability and accuracy of model predictions but also reduces computational load and training time by resizing large images to manageable dimensions. Furthermore, preprocessing can enhance important features and suppress irrelevant variations, ultimately facilitating more effective learning and generalization by the mode [29].

In this study, the original high-resolution images (1689 × 1689 pixels) captured by the optical microscope were resized to multiple target resolutions: 64 × 64, 128 × 128, 224 × 224, 256 × 256, 320 × 320, 448 × 448, 512 × 512, and 600 × 600 pixels. This multi-resolution approach enables systematic evaluation of model performance across different input sizes. Each image was read in color format using OpenCV and converted to RGB to maintain consistency. After resizing, pixel values were normalized to the range to stabilize and accelerate model training.

To further enhance model robustness and reduce overfitting, data augmentation was performed using the ImageDataGenerator. The augmentation techniques included random rotations of up to 10 degrees, width and height shifts of up to 20 % of the image dimensions, shearing transformations with an intensity of up to 0.2 radians (approximately 11.5 degrees) and zooming within a range of 0.8 to 1.2 times the original size (allowing up to 20 % zoom in or out). Additionally, images were randomly flipped horizontally. These augmentation strategies artificially expand the training dataset by introducing variability, enabling the model to learn more generalizable features and improving its performance on unseen data.

Model building

Selecting the appropriate deep learning architecture is critical for balancing classification accuracy, computational efficiency, and inference speed, particularly in industrial defect detection tasks. In this study, we focus on three established CNN models: MobileNetV2, EfficientNetB0, and ResNet152, chosen to represent a spectrum of design philosophies from lightweight, mobile-friendly networks to deeper, high-capacity architectures. This selection enables a comprehensive comparative evaluation across different operational constraints and application needs.

MobileNetV2

MobileNetV2 is renowned for its lightweight design and computational efficiency which make it appropriate for real-time inference and deployment on devices limited in resources. It employs an inverted residual structure with linear bottlenecks and depth-wise separable convolutions, substantially reducing parameters and computational cost while preserving competitive accuracy [30].

EfficientNetB0

EfficientNetB0 is recognized for its balanced approach to scaling model width, depth and resolution through a compound scaling method. This design allows EfficientNetB0 to achieve high accuracy with moderate computational requirements, making it an attractive choice for scenarios demanding both performance and efficiency. EfficientNet models have set benchmarks for accuracy per parameter and FLOP, and EfficientNetB0 is widely adopted for its strong trade-off between accuracy and resource consumption [31].

ResNet152

ResNet152 is distinguished by its deep architecture, featuring 152 layers with residual connections that facilitate the training of very deep networks. It is renowned for achieving cutting-edge accuracy in numerous image classification challenges, particularly when large model capacity is needed to capture complex patterns in extensive datasets. However, this comes at the cost of increased computational making ResNet152 is most suitable for applications where accuracy is paramount and computational

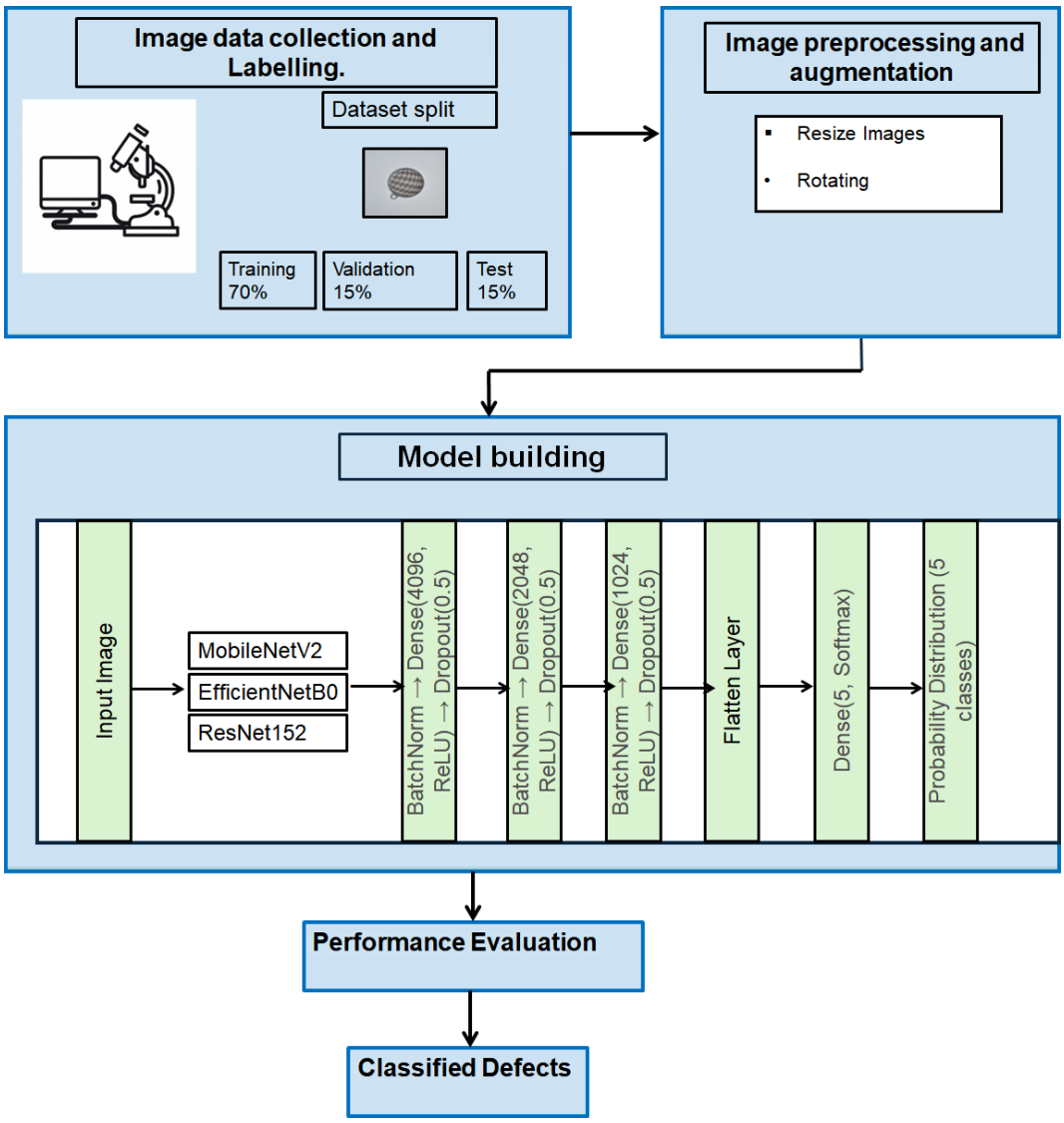

Fig. 2. Image data collection and labelling and memory usage across varying image resolutions. Key findings are discussed.

resources are sufficient [32].

To ensure a fair and systematic comparison, we adopted a backbone swapping strategy, where the same overall model architecture and training pipeline were used, but the feature extraction backbone was varied among MobileNetV2, EfficientNetB0, and ResNet152. In this approach, the backbone model (pretrained on ImageNet) was inserted as the base feature extractor, followed by an identical custom classification head for all experiments. All other components of the workflow, including data preprocessing, augmentation, classifier head structure, training hyperparameters, and evaluation metrics—were kept constant. This methodology isolates the impact of the backbone architecture on classification performance, allowing for direct and unbiased comparison of model efficacy [30, 32].

Performance evaluation

After training, the models are evaluated using the test dataset to assess their classification performance. The evaluation metrics determine how effectively each model identifies and classifies defects, culminating in the final output of classified defect categories.

3. Results and discussion

3.1 Results

This section presents the experimental results obtained from evaluating three deep learning models—MobileNetV2, EfficientNetB0, and ResNet152—for defect classification in water transfer printing manufacturing. The results are analysed in terms of accuracy, inference time, and memory usage across varying image resolutions. Key findings are discussed in the context of their implications for real-world applications, with a focus on balancing performance and computational efficiency. A summary of model performance metrics is provided in Table 2.

Table 2. Performance metrics of MobileNetV2, EfficientNetB0, and ResNet152 across various image resolutions.

Model	Training accuracy (%)	Validation accuracy (%)	Test accuracy (%)	Average inference time per image (ms)
MobileNetV2				
64 x 64	52.60	50.00	50.22	9.588
128 x 128	64.18	61.26	62.78	22.801
224 x 224	81.27	72.52	77.13	54.726
256 x 256	83.97	74.77	73.09	90.027
320 x 320	78.08	67.11	68.60	107.531
448 x 448	70.94	68.46	65.02	234.204
512 x 512	67.76	64.86	64.12	342.571
600 x 600	65.25	58.55	56.95	572.190
Efficient NetB0				
64 x 64	63.22	61.26	57.40	14.02
128 x 128	76.83	70.27	73.54	65.00
224 x 224	86.87	82.43	82.51	94.27
256 x 256	88.32	81.08	82.96	127.64
320 x 320	91.22	83.33	85.20	208.08
448 x 448	94.02	89.64	87.44	417.95
512 x 512	94.98	88.74	87.89	560.98
600 x 600	95.46	88.74	87.89	823.26
ResNet152				
64 x 64	82.72	77.47	78.92	55.91
128 x 128	92.76	83.78	85.65	152.94
224 x 224	97.50	92.34	93.72	417.71
256 x 256	97.97	94.59	92.83	540.45
320 x 320	98.06	94.59	92.83	826.88
448 x 448	98.84	92.8	96.41	2006.19
512 x 512	98.74	95.50	95.52	2111.46
600 x 600	98.46	95.94	94.62	2979.75

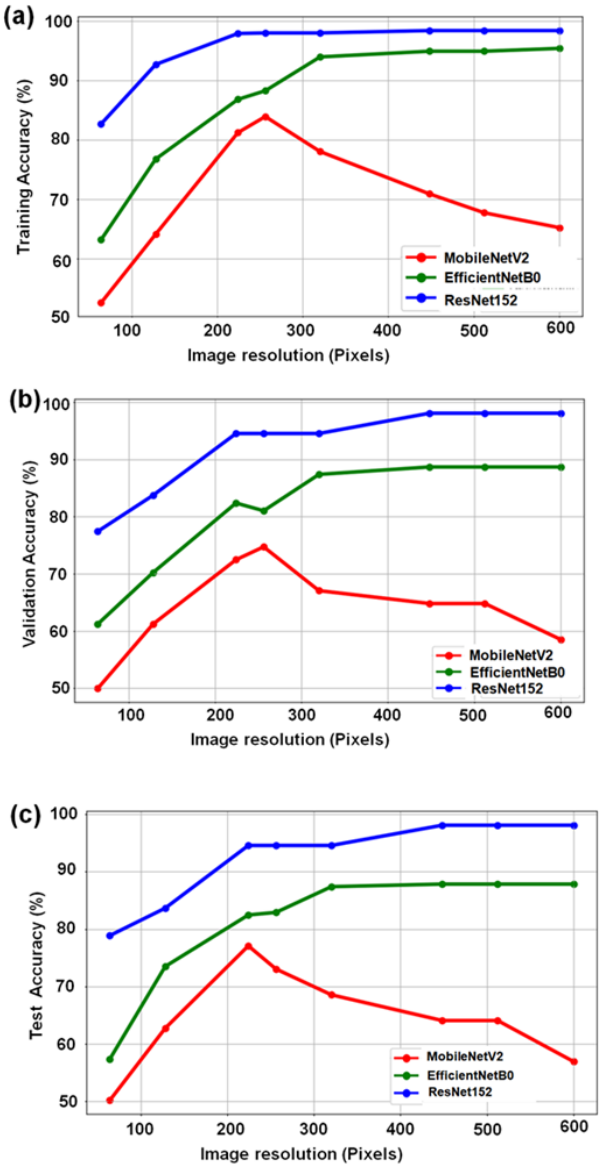

Fig. 3. Accuracy vs resolutions: (a) training accuracy vs image resolution; (b) validation accuracy vs image resolution; (c) test accuracy vs image resolution.

3.1.1 Accuracy vs resolution

Fig. 3 illustrates the relationship between image resolution and accuracy for all three models. Each model is denoted by a distinct color: red for MobileNetV2, green for EfficientNetB0, and blue for ResNet152 to provide clear visual differentiation across the plots. The figure is divided into three subplots: a) training accuracy vs image resolution, b) validation accuracy vs image resolution, and c) test accuracy vs image resolution.

The training accuracy plot demonstrates how each model performance scales with increasing image resolution. MobileNetV2 shows a peak in training accuracy at 256 x 256 resolution (83.97 %), followed by a gradual decline as resolution increases further. EfficientNetB0 exhibits steady improvement in training accuracy as resolution increases, reaching its highest value of 95.46 % at 600 x 600 pixels. ResNet152 consistently achieves the highest training accuracy across all resolutions, peaking at 448 x 448 with 98.84 % before slightly declining at higher resolutions. This trend highlights the superior capacity of ResNet152 to learn from high-resolution images during training.

The validation accuracy plot reflects the generalization performance of the models across different resolutions. MobileNetV2 achieves its highest validation accuracy at 256 x 256 (74.77 %), but its performance declines significantly at higher resolutions, reaching only 58.55 % at 600 x 600 resolution. EfficientNetB0 displays a smoother improvement in validation accuracy as resolution increases, stabilizing at 512 x 512 and 600 x 600 pixels (88.74 %). ResNet152 outperforms both models across all resolutions, achieving its peak validation accuracy at 448 x 448 pixels (95.84 %). These results suggest that ResNet152 maintains robust generalization capabilities even with higher-resolution images.

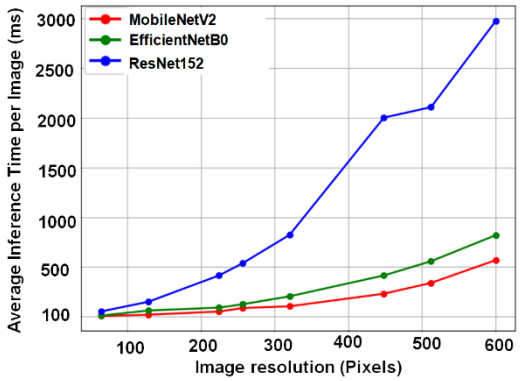

Fig. 4. Inference time vs resolution for three neural network architectures.

The test accuracy plot evaluates the model performance on unseen data across varying resolutions. MobileNetV2 achieves its highest test accuracy at 224 x 224 resolution (77.13 %) but exhibits a sharp decline as resolution increases further, dropping to 56.95 % at 600 x 600 resolution. EfficientNetB0 shows consistent improvement in test accuracy with increasing resolution, reaching its maximum value of 87.89 % at both 512 x 512 and 600 x 600 resolutions. ResNet152 again demonstrates superior performance, achieving its peak test accuracy of 96.41 % at 448 x 448 resolution before slightly declining at higher resolutions.

3.1.2 Inference time vs resolution

Fig. 4 shows the average inference time per image (in milliseconds) for three neural network architectures for all resolutions studied. The inference time for MobileNetV2 starts at 9.588 ms for the smallest resolution (64 x 64 pixels) and steadily increases to 572.190 ms for the largest resolution (600 x 600 pixels). EfficientNetB0 exhibits higher inference times compared to MobileNetV2, beginning at 14.02 ms for 64 x 64 pixels and reaching 823.26 ms at 600 x 600 pixels. For ResNet152, the inference time is significantly higher than the other two models across all resolutions, starting at 55.91 ms for 64 x 64 pixels and increasing sharply to 2979.75 ms at 600 x 600 pixels. The figure demonstrates that inference time grows as image resolution increases for all three models, with ResNet152 showing the largest values across all resolutions, followed by EfficientNetB0 and MobileNetV2.

3.2 Discussion

3.2.1 Impact of resolution on model accuracy

The relationship between image resolution and model accuracy reveals distinct patterns specific to each architecture design characteristics. MobileNetV2 exhibits peak performance at moderate resolutions (224-256 pixels) followed by accuracy degradation at higher resolutions. This performance curve contradicts the conventional expectation that higher resolutions universally improve classification accuracy. The decline in MobileNetV2 accuracy beyond 256 × 256 pixels is particularly significant, with training accuracy dropping from 83.97 % at peak to approximately 65 % at 600 × 600 pixels. This behavior can be attributed to MobileNetV2 depthwise separable convolutions, which were originally optimized for moderate-resolution inputs around 224 × 224 pixels [30-33].

EfficientNetB0 demonstrates a different pattern, with accuracy consistently improving as resolution increases across all datasets. Despite its original design optimization for 224 × 224 inputs, EfficientNetB0 benefits from higher resolutions in this specific defect classification context. This improvement stems from EfficientNetB0 compound scaling method, which balances network depth, width, and resolution more effectively than MobileNetV2. The model shows steady accuracy gains up to 600 × 600 pixels, reaching 95.46 % training accuracy, 88.74 % validation accuracy, and 87.89 % test accuracy at the highest resolution. This suggests that for the visual features in water transfer printing defects, EfficientNetB0 architecture can effectively leverage the additional information present in higher-resolution images.

ResNet152 demonstrates superior accuracy across all resolutions compared to the other models, reflecting its deep architecture capacity to extract complex features. Interestingly, ResNet152 exhibits peak accuracy at 448 × 448 pixels (98.84 % training, 95.84 % validation, and 96.41 % test accuracy) before slightly declining at higher resolutions. This slight performance reduction at very high resolutions likely results from the increased feature space complexity leading to potential overfitting or diminishing returns in information content relative to the computational complexity. The network ability to maintain exceptional performance across resolutions highlights its robust feature extraction capabilities derived from its residual connections, which allow effective gradient flow even in extremely deep architectures.

3.2.2 Computational efficiency and inference time analysis

The inference time measurements reveal critical performance tradeoffs across the three architectures. As image resolution increases, all models show a predictable increase in computational demands, but with substantially different scaling patterns. MobileNetV2 maintains the lowest inference times across all resolutions, starting at just 9.588 ms for 64 × 64-pixel images and increasing to 572.190 ms at 600 × 600 pixels. This relatively modest increase reflects the model efficient design through depthwise separable convolutions that reduce computational complexity compared to standard convolutions [30-33].

EfficientNetB0 exhibits moderately higher inference times than MobileNetV2, ranging from 14.02 ms at 64 × 64 pixels to 823.26 ms at 600 × 600 pixels. Though designed for efficiency, EfficientNetB0 compound scaling approach results in greater computational demands than MobileNetV2, particularly at higher resolutions where the balanced scaling of network depth, width, and resolution increases overall complexity.

ResNet152 demonstrates significantly higher inference times across all resolutions, with processing times ranging from 55.91 ms at 64 × 64 pixels to 2979.75 ms at 600 × 600 pixels. This substantial computational demand reflects ResNet152 deep architecture with 152 layers, resulting in approximately 5 times longer inference times than MobileNetV2 at comparable resolutions. The exponential increase in inference time for ResNet152 at higher resolutions is particularly notable, limiting its practical application in real-time manufacturing environments where processing speed is critical.

These results confirm a non-linear relationship between image resolution and model performance: as resolution increases from 64 × 64 to 600 × 600 pixels, test accuracy improves by 22–53 % depending on the model, while inference times escalate dramatically—by over 5100 % to nearly 5900 % for all architectures. Specifically, the test accuracy of MobileNetV2 rises by approximately 22 % (from its lowest to peak), EfficientNetB0 by about 47 %, and ResNet152 by around 53 %. However, this substantial gain in accuracy comes at the cost of a dramatic increase in inference time, with MobileNetV2 inference time increasing by nearly 5,870 %, EfficientNetB0 by 5780 %, and ResNet152 by 5,230 % from the lowest to the highest resolution. This underscores the critical trade-off between accuracy and computational efficiency when selecting model architectures and input resolutions for practical deployment [34].

3.2.3 Performance trade-offs and manufacturing implications

Balancing accuracy and computational efficiency

The observed performance characteristics reveal important trade-offs between accuracy and computational efficiency that directly impact practical implementation in manufacturing environments [35]. Considering both metrics together, MobileNetV2 demonstrates optimal efficiency at 224 × 224-pixel resolution, where it achieves 77.13 % test accuracy with relatively low memory requirements (68.78 MB) and moderate inference times. This represents an important balance point where the model maintains acceptable accuracy while minimizing computational demands, making it suitable for integration into resource-constrained production environments or edge devices deployed on manufacturing floors.

EfficientNetB0 offers a middle-ground solution, with memory requirements (75.62 MB) slightly higher than MobileNetV2 but substantially lower than ResNet152. Its optimal performance point occurs at higher resolutions (512 × 512 pixels), where it achieves 87.89 % test accuracy while maintaining more reasonable inference times than ResNet152. This makes EfficientNetB0 well-suited for manufacturing scenarios where higher accuracy is required but real-time processing remains important, such as high-value manufacturing lines where defect detection accuracy justifies moderate computational costs [36].

ResNet152, despite its superior accuracy, presents significant implementation challenges in typical manufacturing environments due to its extensive memory requirements (294.85 MB) and lengthy inference times at higher resolutions. The model optimal operating point occurs at 448 × 448 pixels, where it achieves peak accuracy before diminishing returns at higher resolutions. This architecture would be most suitable for critical quality control applications where detection accuracy is paramount and processing time constraints are less strict, such as final inspection stages or high-precision manufacturing where the cost of missed defects outweighs computational considerations.

Practical implementation in water transfer printing manufacturing

In the specific context of water transfer printing manufacturing, these findings have direct implications for automated defect detection system design. The visual characteristics of the five identified defect categories - entrapped air defects, film creasing defects, film delamination, pattern distortion, uniform transfer, present varying levels of complex detection. Some defects, such as air defects or significant delamination, may be readily identifiable at lower resolutions, while more subtle defects like minor stretching or small bubbles may require higher-resolution imaging for reliable detection.

For high-volume production lines where processing speed is critical, MobileNetV2 at 224 × 256-pixel resolution offers sufficient accuracy for detecting major defects while maintaining real-time processing capabilities. This configuration would be particularly suitable for initial screening processes, where obviously defective items can be quickly identified and removed from the production flow.

For more comprehensive inspection systems where accuracy is prioritized over speed, EfficientNetB0 at 512 × 512-pixel resolution provides a good compromise, offering significantly improved detection accuracy across all defect types with moderate computational requirements. This configuration would be well-suited for secondary inspection stages or manufacturing processes where defect detection accuracy directly impacts product quality and customer satisfaction.

ResNet152 at 448 × 448-pixel resolution represents the highest-accuracy option, potentially suitable for final quality assurance processes or research applications where detailed defect classification is required. While its computational demands limit real-time applications, it could serve as a benchmark system for validating the performance of more efficient models or for offline analysis of challenging defect cases. The implementation strategy should follow a multi-tier inspection approach, commonly used in modern manufacturing systems, where lightweight models like MobileNetV2 handle initial screening, balanced models like EfficientNetB0 perform secondary inspection, and high-accuracy models like ResNet152 are reserved for final quality assurance or challenging cases requiring detailed analysis.

4. Conclusions

In this study, we evaluated the performance of three neural network architectures: MobileNetV2, EfficientNetB0, and ResNet152, for image classification in the manufacturing context of water transfer printing. The analysis revealed clear trade-offs

between accuracy, inference time, and computational efficiency across varying image resolutions. MobileNetV2 demonstrated the lowest inference times, making it ideal for real-time applications on resource-constrained devices, though it achieved moderate accuracy levels. EfficientNetB0 emerged as the most balanced model, providing high accuracy (~90 %) while maintaining manageable inference times, making it suitable for production lines requiring both speed and precision. ResNet152 achieved the highest accuracy (~90 %–95 %) but at the cost of significantly higher inference times and computational demands, limiting its applicability to high-stakes scenarios where precision is paramount. The trends in accuracy across training, validation, and test datasets confirmed that higher resolutions improve classification performance due to better feature extraction, though this comes with increased computational costs. The trade-off analysis highlighted that EfficientNetB0 offers the best balance between accuracy and efficiency for water transfer printing classification tasks. These findings provide valuable insights into model selection for manufacturing applications, emphasizing the importance of aligning computational resources with task-specific requirements to optimize performance and operational efficiency.

Acknowledgments

This work was supported by the Basic Science Research Program through the National Research Foundation of Korea (NRF) funded by the Ministry of Education (NRF-2021R1I1A3059714). This work was also supported by a project for Collabo R&D between Industry, University, and Research Institute funded by Korea Ministry of SMEs and Startups in 2025 (RS-2025-02313349). This research was also supported by the Ministry of Trade, Industry & Energy (MOTIE) and the Korea Institute for Advancement of Technology (KIAT) of the Republic of Korea (RS-2025-17022968). This work was also supported by the Ministry of Science and ICT (MSIT) of the Republic of Korea, through the "Digital Innovation Hub Sup-port Project" funded by the National IT Industry Promotion Agency (NIPA) and the Chungbuk Innovation Institute of Science & Technology (CBIST).

Conflict of interest

The authors declare no competing interests.

References

[1] H. Zhou et al., Transfer printing and its applications in flexible electronic devices, *Nanomaterials*, 9 (2) (2019) 283.

[2] R. Selmi et al., Organic and metallic sensors on complex 3-D object using an original method: water transfer printing, *IEEE Sensors Letters*, 7 (9) (2023) 1-4.

[3] S. Cai et al., Three-dimensional triboelectric nanogenerator manufacturing using water transfer printing, *Chemical Engineering Journal*, 493 (2024) 152862.

[4] D.-H. Kim et al., Dissolvable films of silk fibroin for ultrathin conformal bio-integrated electronics, *Nature Materials*, 9 (6) (2010) 511-517.

[5] W. Yeo et al., Multifunctional epidermal electronics printed directly onto the skin, *Advanced Materials*, 25 (20) (2013) 2773-2778.

[6] B. Le Borgne et al., Water transfer printing enhanced by water-induced pattern expansion: toward large-area 3D electronics, *Advanced Materials Technologies*, 4 (4) (2019).

[7] Y. Feng, W. Hong and X. Chen, Eco-friendly water transfer printing free of primers and activators, *ACS Applied Polymer Materials*, 3 (7) (2021) 3569-3575.

[8] J. Liu et al., Sacrificial layer-assisted nanoscale transfer printing, *Microsystems & Nanoengineering*, 6 (1) (2020) 80.

[9] M. M. Rahman Khan, Md. M. H. Rumon and M. Islam, Synthesis, rheology, morphology, and mechanical properties of biodegradable PVA-based composite films: a review on recent progress, *Processes*, 12 (12) (2024) 2880.

[10] M. Harnois et al., An improved fabrication technique for the 3-D frequency selective surface based on water transfer printing technology, *Scientific Reports*, 10 (1) (2020) 1714.

[11] Z. Chen, C. Zhang and Z. Zheng, Advancements in transfer printing techniques for flexible electronics: adjusting interfaces and promoting versatility, *International Journal of Extreme Manufacturing*, 6 (5) (2024) 052005.

[12] A. Gurusekaran, A. Elouadi and A. Mansour, Analysis of dispense and water transfer printing as fabrication methods for UHF antennas on 3D printed substrates, *2022 IEEE Electron Devices Technology & Manufacturing Conference (EDTM)*, Oita, Japan (2022) 110-112.

[13] Y. Zhang et al., Computational hydrographic printing, *ACM Transactions on Graphics*, 34 (4) (2015) 1-11.

[14] W. Chen et al., A comparative study of vehicle frame load spectrum extraction using SA-1DCNN deep learning and virtual iteration, *Journal of Mechanical Science and Technology*, 38 (10) (2024) 5297-5309.

[15] S.-H. Kim et al., Development of machine learning models for material classification and prediction of mechanical properties of FDM 3D printing outputs, *Journal of Mechanical Science and Technology*, 39 (2) (2025) 541-552.

[16] L. Alzubaidi et al., Review of deep learning: concepts, CNN architectures, challenges, applications, future directions, *Journal of Big Data*, 8 (1) (2021) 53.

[17] M. Ge et al., Deep learning analysis on microscopic imaging in materials science, *Materials Today Nano*, 11 (2020) 100087.

[18] K. Choudhary et al., Recent advances and applications of deep learning methods in materials science, *NPJ Computational Materials*, 8 (1) (2022) 59.

[19] M. Raisul Islam et al., Deep learning and computer vision techniques for enhanced quality control in manufacturing processes, *IEEE Access*, 12 (2024) 121449-121479.

[20] A. Ettalibi, A. Elouadi and A. Mansour, AI and computer vision-based real-time quality control: a review of industrial applications, *Procedia Computer Science*, 231 (2024) 212-220.

[21] S. M. Azimi et al., Advanced steel microstructural classification by deep learning methods, *Scientific Reports*, 8 (1) (2018) 2128.

[22] E. Kim et al., Deep learning-based automatic defect classification for semiconductor manufacturing, J. C. Robinson and M. J. Sendelbach (Eds.), *Metrology, Inspection, and Process Control XXXVII*, 12496 (2023) 49.
[23] S. Samsi et al., From words to watts: benchmarking the energy costs of large language model inference, *2023 IEEE High Performance Extreme Computing Conference (HPEC)*, Boston, USA (2023) 1-9.
[24] K. De and M. Pedersen, Impact of colour on robustness of deep neural networks, *2021 IEEE/CVF International Conference on Computer Vision Workshops (ICCVW)*, Montreal, Canada (2021) 21-30.
[25] V. Thambawita et al., Impact of image resolution on deep learning performance in endoscopy image classification: an experimental study using a large dataset of endoscopic images, *Diagnostics,* 11 (12) (2021) 2183.
[26] H. R. Sheikh and A. C. Bovik, Image information and visual quality, *IEEE Transactions on Image Processing,* 15 (2) (2006) 430-444.
[27] S. Tang et al., The effect of image resolution on convolutional neural networks in breast ultrasound, *Heliyon,* 9 (8) (2023) e19253.
[28] C. F. Sabottke and B. M. Spieler, The effect of image resolution on deep learning in radiography, *Radiology: Artificial Intelligence,* 2 (1) (2020) e190015.
[29] C. Shorten and T. M. Khoshgoftaar, A survey on image data augmentation for deep learning, *Journal of Big Data,* 6 (1) (2019) 60.
[30] M. Sandler et al., MobileNetV2: inverted residuals and linear bottlenecks, *2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition*, Salt Lake City, USA (2018) 4510-4520.
[31] S. Torne et al., VGG-16, VGG-16 with random forest, ResNet50 with SVM, and EfficientNetB0 with XGBoost-enhancing bone fracture classification in X-ray using deep learning models, *IEEE Access,* 13 (2025) 25568-25577.
[32] M. Shafiq and Z. Gu, Deep residual learning for image recognition: a survey, *Applied Sciences,* 12 (18) (2022) 8972.
[33] V. Kamath and A. Renuka, Deep learning-based object detection for resource constrained devices: systematic review, future trends and challenges ahead, *Neurocomputing,* 531 (2023) 34-60.
[34] H. Kaur, R. Sharma and J. Kaur, Comparison of deep transfer learning models for classification of cervical cancer from pap smear images, *Scientific Reports,* 15 (1) (2025) 3945.
[35] L. Bai and W. H. Xu, Improved printed circuit board defect detection scheme, *Scientific Reports,* 15 (1) (2025) 2389.
[36] Y. Xu et al., The steel surface multiple defect detection and size measurement system based on improved YOLOv5, *Journal of Electrical and Computer Engineering,* 2023 (2023) 1-16.

Joel Ndikumana is currently a Postdoctoral Researcher in the Department of Mechanical Engineering at Sejong University, South Korea. He holds a Ph.D. and a master's degree in electronic engineering from Hallym University, South Korea. His research focuses on semiconductor devices, particularly OTFTs and OLEDs, with expertise in thin film processes, micro- and nanofabrication techniques, and the application of AI in manufacturing. Prior to his current position, he worked as a Postdoctoral Researcher and Lecturer in mechatronics at Konkuk University and as a Research Assistant at Hallym University.

Gyuseok Lee had his undergraduate degree from Konkuk University Glocal campus, Republic of Korea in 2025. His research interests are focused on Autonomous fabrication system and convolutional neural network design for industrial process optimization.

Kunsik An is an Assistant Professor in the Department of Mechanical Engineering at Sejong University, South Korea. Previously, he was at Konkuk University Glocal Campus and worked as a Postdoctoral Researcher at KITECH. He holds a Ph.D. from Seoul National University and degrees from KAIST. His research focuses on manufacturing AI, electronic devices (TFT, OLED, QLED), and printed electronics. His notable achievements include image-based impedance spectroscopy and nanostructures for quantum dot LEDs.